

SYNAPSE

Master Plan v2 — The Full Platform

Learners · Teachers · Parents · Schools

Every major learning app solves one slice. Synapse connects all the slices — and adds the one thing none of them can: the classroom itself.

Levels: Primary · Junior Secondary · Senior Secondary → Colleges & Tertiary later

Covers and beats: Notion · NotebookLM · Gauth · ChatEdu · SchoolGo — plus school branding, study pathways, and the teacher-parent-school loop.

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1. The Vision

Synapse is the connected learning platform where learners improve, teachers teach instead of doing paperwork, parents finally see the truth, and schools run on evidence. It is lightweight in feel — simple pages — but the pages are wired together, and that wiring is the product.

Principles

- Synapses over pages. Every page reads from and writes to the shared Learner Model. Data flows; nobody re-enters anything; each feature makes the others smarter.
- The curriculum is ground truth. Digitized Zambian syllabi (ZECF 2023, CDC codes like 4.1.4.1). Every plan, question, answer, artifact and report cites it.
- The knowledge base grows itself. Any uploaded paper, textbook or note is auto-digitized, tagged to competences, and entered into the bank. New ECZ papers are known the day after upload — forever.
- Study pathways, not study sessions. Learning is continuous: a pathway follows the learner across pages, terms and years — resumable anywhere, even offline (referring to notes in church, revising on the bus).
- Schools are brands. Each school gets a branded account: logo and colors on the dashboard, lesson plans, schemes of work, tests, report cards and certificates. Documents look like the school's, not generic.
- Offline-first, language-inclusive. Everything works without signal and syncs later. Explanations and audio in English plus major Zambian languages.
- Privacy by design. Learners are minors: permissions asked first, monitoring opt-in and parent-controlled, guardians see only their children, on-device processing for anything sensitive.

2. Architecture — One Brain, Four Worlds

At the core: the Curriculum Engine (subjects → grades → terms → topics → sub-topics → competences with official codes) and the Knowledge Bank (past papers, questions, solutions, materials — auto-digitized). Around them, four worlds connected by synapses:

World	Feeds into Synapse	Receives from Synapse
Learner	Study sessions, practice & exam results, artifacts, focus time	Adaptive pathway, assignments, resources, study packs
Teacher	Plans, schemes, tests, mastery scores, upload verifications	Class learner models, review queue, curriculum & paper bank
Parent	Guardian link, viewing activity	Verified progress, study reports, alerts, messages, meetings
School	Structure, branding, announcements, enrollment	Aggregate analytics, document branding, Ministry-ready evidence

One engine serves Primary, Junior Secondary and Senior Secondary — each level is syllabus data in the same tables. Colleges and tertiary join later through the same ingestion pipeline.

3. The Learner World — in Detail

Ten connected spaces. Each is useful alone; together they form a study pathway that never loses the thread.

3.1 Home (Today)

- Auto-built daily plan; streaks and weekly goals; flashcards due today (spaced repetition); exam countdowns; weak-topic nudges ("sets troubled you — 10 minutes?"); pick up where you left off — any page, any device.
- Optional mood/energy check-in — Synapse schedules hard topics when the learner says they feel sharp.

3.2 Synapse AI (the engine room)

- Curriculum-grounded chat with citations ("this is Commerce 4.1.4, ZECF 2023") — ChatEdu beaten by being official.
- Snap & Solve — photograph any problem; symbolic math engine solves, AI explains every step (Gauth beaten by reliability).
- Voice Tutor — talk it through, hear the explanation back; hands-free revision for low-bandwidth homes.
- Local-language explanations — English, Bemba, Nyanja and more; ask in one language, revise in another.
- Artifact generator — "make me a 3D model of the heart", "a projectile simulation", "an interactive timeline of the Zambian economy". Sandboxed interactive HTML, saved to Library, shared to study groups, tagged to a competence so classmates on that topic find it automatically.
- Modes: Explain · Guide · Solve · Practise · Assess · Create · Organise · Plan. Eight verbs, one engine.

3.3 Library (Notion × NotebookLM)

- Files, notes, folders; PDF reader with highlight and annotate; document scan (OCR of handwritten notes); web clipper; Ask My Library — answers grounded in your own materials with citations to the exact page.
- Auto-generated companions: every note gets a summary, key points, flashcards and a mini-quiz (Break It Down).
- Formula sheets & glossaries per subject and grade, generated from the syllabus and kept updated.

3.4 Practice (My Circle)

- Five-step loop — Capture → Understand → Practise → Review → Master — with weekly completion. Study in loops, not lines.
- ECZ paper bank — real papers segmented question-by-question, each tagged to a competence, with worked solutions and examiner marking points.
- Paper Mode — open a past paper; tap any question for its worked solution, marking guide and an instant mini-quiz on that topic. A static PDF becomes a tutor.
- Personalized mocks — "I've covered algebra and sets" → authentic ECZ-format timed exam from real bank questions, fresh per-learner number variants, auto-marked with explanations.
- Exam Simulator — full papers under real timing; adaptive selection weights your needs-work competences.

- Drills & flashcards with spaced repetition scheduled automatically (day after → three days → a week).

3.5 Study Groups

- School-scoped groups: shared notes and flashcard decks, group quizzes, challenges with leaderboards (opt-in), discussion threads.
- Escalate any question to Synapse AI — the answer returns to the whole group. One member's upload becomes everyone's study pack.

3.6 Planner

- Tasks, daily goals, focus timer; revision plan generator — "ECZ Maths in 21 days" becomes a paced daily plan.
- The planner schedules weak-topic practice and flashcard reviews automatically from the Learner Model — it plans around your reality, not just your to-do list.

3.7 Progress (outcomes)

- Mastery per subject and competence, strengths and needs-work, term-over-term trends, exam-readiness estimate.
- Career lens: "where does this competence lead?" — biology competences link to nursing, maths to engineering, commerce to accounting — motivation through direction.
- Certificates: milestone certificates (school-branded) for completed topics and mastered terms.

3.8 Learn Anywhere (offline & audio)

- Offline-first: notes, flashcards, papers and downloaded lessons work with no signal; everything syncs when back online.
- Audio lessons ("revision radio"): any note or topic becomes a spoken summary — revise on the bus, in church before service, while cooking.
- Cross-device pathway: phone at school, parent's tablet at home — the pathway resumes exactly where it stopped.

3.9 Achievements

- Badges and milestones (first perfect quiz, 7-day streak, topic mastered), class leaderboards (opt-in), term certificates.

3.10 Me (Profile & Family Link)

- Profile, joined classes, language preferences, Family Link (guardian connects with the learner code), notification controls, data/privacy settings.

4. The Teacher World — in Detail

Fourteen connected capabilities — an AI workspace plus a classroom command center. Everything a Zambian teacher needs, nothing they have to format themselves.

4.1 AI Lesson Planning

- Cascading curriculum dropdowns (grade → subject → topic → sub-topic → competence — official data, never free text).
- Full MoE-format generation: general/specific competences, lesson goal, rationale, prior knowledge, references (ZECF 2023), learning environment, low-resource-aware materials, expected standard, and the 5-stage progression table (Teacher activity / Learner activity / Assessment criteria).
- Edit anything inline; regenerate any section; version history keeps every draft.

4.2 Schemes of Work

- Fill Week 1 — the engine continues in syllabus order across the term; weeks can be marked Revision or Test; full MoE table generated.

4.3 Voice-to-Plan

- Describe the lesson aloud in English or a local language — Synapse structures it into the official format. For teachers faster at speaking than typing.

4.4 Tests & Exam Generation

- Quick tests per competence, or full ECZ-format exams learned from the real paper bank — authentic structure, command words, mark distribution — with complete marking guides and per-student variants.
- Diagram support: LaTeX-rendered math, real music notation, generated figures for technical subjects.

4.5 Classroom & Scores

- Create class → class code; learners join in seconds. Mastery entry after each lesson: mastered yes/no per learner in under two minutes.
- Light attendance; class list with guardian-link status (who is connected to parents, who isn't).

4.6 Assignments & AI Marking

- Assign a paper or generated exam → lands in every student's Planner with due date and notification. Submissions AI-marked against the guide; teacher verifies with one-tap overrides.

4.7 Class Analytics

- Mastery heatmap per competence; "who needs what" list per learner; trends across the term; comparison to the class average — all feeding the adaptive engine.

4.8 Adaptive Engine

- Below-threshold mastery → next plan opens with remediation; strong mastery → enrichment. The class teaches the plan, and the plan adapts to the class.

4.9 Branded Documents

- Every plan, scheme, test, report card and certificate carries the school logo, school colors, teacher name and school header — print-ready PDFs that look official for inspectors and the Ministry.

4.10 Teacher Library

- School-scoped shared repository: reuse colleagues' plans, contribute your own, rate and organize by topic. A school's collective knowledge stops leaving when a teacher transfers.

4.11 Report Cards & Comments

- Term report cards auto-composed from mastery data — school-branded, per learner, with AI-drafted teacher comments editable in one tap.

4.12 Communication

- Announcements to a class or the whole school; parent messaging with read receipts; parents request meetings from teacher-published slots.

4.13 Review Queue & Professional Growth

- Uploaded materials with low auto-tagging confidence wait here — one tap confirms or fixes; teachers curate the platform in seconds.
- Lesson observation notes and teaching-standards tracking begin continuous professional development inside the workspace.

4.14 Calendar & Deadlines

- Term planner: schemes, tests, assignments and events on one calendar; deadline reminders to learners and parents automatically.

5. The Parent World — in Detail

- Family Link & multi-child: connect with learner codes; support several children in one account, switching between them instantly.
- Progress dashboard: mastery per subject and competence, trends term over term, exam-readiness, class-average comparison — the truth, live, not end-of-term surprises.
- Verified study reports: "Elena studied 32 minutes today — engaged, verified". Session logs from the focus system; optional on-device learning confirmation (opt-in, parent-controlled).
- Alerts that matter: missed assignments, slipping mastery, upcoming tests, absence notes — tuned to urgency, not spam.
- Documents inbox: school-branded report cards, test results and certificates arrive as they are published.
- Messaging & meetings: direct chat with teachers and the school; book meetings from teacher-published slots; announcements feed.
- Home support: "tonight, ask your child about osmosis" — conversation starters matched to this week's syllabus, so parents help without knowing the content.
- Language & access: plain-language summaries in English and major Zambian languages; (later) USSD result checks for parents without smartphones.

6. The School World — in Detail

- Branded account: upload logo, set colors, school name and details — branding flows to the dashboard, all generated documents, report cards, certificates and parent-facing views. The platform looks like the school.
- Codes & onboarding: school code; staff codes for teachers; class codes for learners; learner codes for guardians — four codes connect an entire school in one afternoon.
- Oversight dashboard: every teacher, class, grade and subject — planning activity, mastery heatmaps, comparative trends, engagement. Weak-competence view across the whole school.
- Enrollment management: learners, classes, staff directory and roles, guardian-link status — one source of truth, no spreadsheets.
- Announcements & events: targeted broadcasts (one grade, whole school, parents only); events calendar (exams, open days, sports).
- Report cards & certificates: generate the whole term's reports at once — school-branded PDFs; certificates for achievements.
- Academic analytics: which competences the school struggles with per grade and subject; intervention tracking; evidence ready for the Ministry and inspectors.
- Resource library: school policies, circulars, schemes archive — searchable, always current.
- Subscriptions & tiers: per-school plans (learners/parents free; schools pay for management depth and branding).
- District & Ministry rollup (later): anonymized aggregates give the Ministry a live national picture it has never had.

7. The Synapses — How It All Works Together

Signal	Flow through Synapse
Learner studies	Focus session + practice results → Learner Model → Progress (learner) · adaptive plan (teacher) · verified report (parent)
Teacher teaches	Plan delivered → mastery scored → class mastery → next plan auto-adjusts → parent sees outcome
Upload anything	PDF (paper, textbook, notes) → auto-digitized → competence-tagged → bank grows → exams get better for everyone
Artifact created	"3D heart model" → tagged to competence → surfaced to classmates on that topic → shared to study group
Assignment set	Teacher assigns → student Planner + alert → AI marking → mastery update → parent notified
New exams arrive	2027 ECZ paper uploaded → tagged overnight → fresher prep for every learner
Report time	Mastery data → branded report cards → parent inbox → school archive → Ministry evidence
School acts	Weak school-wide competence found → announcement + intervention → teacher plans reflect it
Pathway continuity	Church, bus, home — offline notes and audio keep the pathway alive; sync resumes the thread anywhere
Branding everywhere	School logo set once → appears on every document, report, certificate and parent view automatically

8. Study Pathways — the Product Philosophy

- A pathway is the learner's continuous route through the curriculum: where they are, what's next, what's weak, where it leads (careers).
- It never resets: terms roll over, weak competences resurface for review, new topics connect to prior ones automatically.
- It lives everywhere: every page advances it; offline time counts; audio revision counts; a teacher's lesson moves it; a parent's evening question reinforces it.
- Study pathways, not study apps — the difference between a tool and a journey.

9. Trust, Integrity & Anti-Cheat

- Permissions first at install (focus/monitoring) — explicit, revocable, parent-controlled. Nothing covert, ever.
- Marking guides separate from questions; revealed only after submission; questions in live assignments locked until due date; every solution view logged and auditable by teachers.
- Per-learner randomized variants defeat answer-key sharing; high-stakes prep uses fresh AI-written parallel questions.
- Database-level security: guardians see only their children; vision processing on-device; no raw images stored or transmitted.

10. The MVP — Real, Not Ambitious

The smallest platform that proves the full loop, with the breadth that makes each stakeholder stay.

#	Component	In the MVP	Deliberately later
1	Curriculum engine	3 subjects fully digitized (CS F3, Commerce F4, Maths G12)	Full national coverage; more levels
2	Teacher world	Plans, schemes, class codes, scores, adaptive adjustment, test generation	Voice-to-plan, observations, shared library
3	Learner world	Join class, assignments, grounded AI chat, flashcards, Progress	Artifacts gallery, study groups, audio radio
4	Paper bank	Ingest 3 years of ECZ papers for MVP subjects; auto-tagging + teacher review queue	Self-serve uploads open to all users
5	Personalized mocks	Adaptive selection from bank; per-learner variants; auto-marking	Full exam simulator depth
6	Parent world	Family Link, progress dashboard, verified study-time reports, alerts	Meetings booking, home-support prompts, USSD
7	Focus shield	Study-block notification blocking (Android)	Full app-lock depth; vision verification
8	School world	Branded account (logo on documents), codes onboarding, oversight dashboard	Analytics depth, timetables, attendance, subscriptions
9	Offline mode	Notes, flashcards, papers cached; sync on reconnect	Full offline lesson playback
10	Languages	English UI; AI explanations in Bemba/Nyanja	Full UI localization

Build order

- Wks 1-4: schema live; first syllabus digitized; first perfect lesson plan (matches the MoE format exactly).
- Wks 5-8: class codes + scores + adaptive engine; paper pipeline on real ECZ PDFs; school branding (logo → documents).
- Wks 9-12: learner core + Family Link + mocks + focus shield; pilot in 2 schools, one per level band.

Pilot success criteria

- Teachers save ≥ 3 planning hours/week; $\geq 70\%$ weekly active use for a full term.
- ≥ 1 guardian linked per learner, viewing weekly; parents rate visibility better than any previous channel.
- Schools use branded documents in real meetings/inspections; ≥ 3 years of papers per subject in the bank.

11. Technical Foundation

- Flutter (Android-first) · Supabase (Postgres + RLS + storage) · LLM API behind serverless functions · symbolic math engine · sandboxed HTML artifacts · offline sync layer · TTS audio pipelines.
- Data foundation: synapse_schema.sql (v0.2) + papers migration (v0.3) + artifacts/exams migration (v0.4).
- Open dependency: real source documents — one CDC syllabus PDF and ECZ past papers to run the first end-to-end ingestion.

Immediate next step: digitize one syllabus and ingest one past paper — real content in, structured knowledge out.